

Random Access and its relation to Compressed Sensing, Heavy Hitter, & Group Testing

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Based on

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Group
Testing

Background

2^n test $\sim 10^6$
10000 $\sim 10^4$

IDEA

Mathematical
formulation

$x \in \{0,1\}^n$
 $y_i = \sum_{j \in S_i} x_j$ if phone j is sick
 $A \in \{0,1\}^{m \times n}$
 $y = Ax$ "sick status"

Example

$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$

What can A do?

What if there are 22 bugs?

Heavy
Hitter

What are the
most requested
pages/sites/webpages/...

Key: which number
"is" the largest
and how?

Other count
1/1000000
1/1000000

Actually, not a good idea
These numbers are spread out in time
We are looking for a few items
causing / causing / causing

IDEA

$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ "sick status"

Fold x by A
 $y = Ax$
For some phone
The idea is like that?

Compressed
Sensing

reduce
 $x \rightarrow y$

clear
picture

Mathematical Model

$y = Ax \in \mathbb{C}^{m \times n}$

$x = x \in \mathbb{C}^{n \times 1}$

$y = Ax \in \mathbb{C}^{m \times 1}$

solve x

Caution: "Someone is too large
the other one how large?"
($k \ll n \ll m$)

$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \in \mathbb{C}^m$

$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{C}^n$

Technically, this will work
if A is "random",
i.e. $\|Ax\| = \sqrt{m} \|x\|$

Why? Because if x is 0 or x
then $\|Ax\| = 0$ or $\|Ax\| = \sqrt{m} \|x\|$
then $\|y\| = \|Ax\|$ is 0 or $\sqrt{m} \|x\|$

And then there are
people trying to construct
counterexamples.....

$A = \begin{bmatrix} 1 & 1 & 1 & \dots \\ 1 & 1 & 1 & \dots \\ 1 & 1 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$

$A = \begin{bmatrix} 1 & 1 & 1 & \dots \\ 1 & 1 & 1 & \dots \\ 1 & 1 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$

Multiple
Access
Channel

$y := \sum_j e^{s_j + i\phi_j} \cdot M_j$

s_j : strength of signal

ϕ_j : phase / sync error

M_j : message

Caution: even larger n
 n is # possible messages

Caution: we don't know who is active/who is idle

Caution: If two signals collide they destroy each other

Challenges

Error Model

Noiseless Group Testing

$0+0=0$

$0+1=1$

$1+0=1$

$1+1=1$ not 2

(Good for modeling interference)

Noisy Group Testing:

$\begin{matrix} 0 \\ 1 \end{matrix} \rightarrow \begin{matrix} 0 \\ 1 \end{matrix}$ BSC

$\begin{matrix} 0 \\ 1 \end{matrix} \rightarrow \begin{matrix} 0 \\ 1 \end{matrix}$ AWGN

Heavy Hitter:

$\begin{matrix} 0 \\ 1 \\ 2 \end{matrix} \rightarrow \begin{matrix} 0 \\ 1 \\ 2 \end{matrix}$ asymmetric channel

Compressed Sensing:

$y = Ax + \text{noise}$ additive

But $y = \begin{bmatrix} 1 & \epsilon & \epsilon & \epsilon \\ \epsilon & 1 & \epsilon & \epsilon \\ \epsilon & \epsilon & 1 & \epsilon \\ \epsilon & \epsilon & \epsilon & 1 \end{bmatrix} Ax$

multiplicative noise

some noise in each composition
($1 + \epsilon \cos(\theta)$) $\rightarrow 1$

Recently:

Generalized Group Testing

Pick a function $f: \mathcal{N} \rightarrow [0,1]$ (monotonic)

If input is x , output is 0 w.p. $1 - f(x)$

1 w.p. $f(x)$

Decoding Complexity

(aka, what's your parameter regime?)

n : # all possible signals

k : # active signals

m : # measurements

$m \sim k \log n$ for GT/HH/CS

But decoding complexity $\sim kn$

Latest: $k \text{ poly}(\log(n))$ complexity

if $m \sim k \text{ poly}(\log(n))$

Technique: Break the signal

phone 1 $\square$: $\begin{matrix} \text{signal} \\ \text{noise} \end{matrix}$

phone 2 $\square$: $\begin{matrix} \text{signal} \\ \text{noise} \end{matrix}$

Receiver $\triangle$: $\begin{matrix} \text{signal} \\ \text{noise} \end{matrix}$

Each signal piece should contain phone ID

$\Rightarrow$ phone ID is $\log n$ long

$\Rightarrow$ each piece is long

$\Rightarrow$ few pieces

$\Rightarrow$ high collision prob

$\Rightarrow$ bad

Isolation

(which is simpler than Group Testing)

★ I don't want to identify the signals ★

★ I just want to separate them into bags ★

★ such that each bag contains ≤ 1 signal ★

$\begin{matrix} * \\ * \\ * \end{matrix}$ $\rightarrow$ $\begin{matrix} * \\ * \\ * \end{matrix}$

Doing it blindly $\rightarrow$ birthday paradox

So, "measure it" $\triangle$

But minimize #measurements

minimize #sequential measurement

(P.S. measurement is noisy)